Physics 525: Plasma Wakefield Acceleration

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1 Introduction

Wakefield particle acceleration is an exciting application of plasma physics, which promises accelerating gradients 10-1000 times greater than that of a traditional RF cavity. Understanding it is key to creating more economical, compact, and higher energy particle accelerators for physics research in the future. The technique is to shoot a beam of ions through a plasma called a drive beam which creates electric wakefields. We can then place a witness beam behind the drive beam to be accelerated by those wakefields. The next effect is that energy is transferred from the drive beam to the witness beam. This LaTeX report describes the use of the code in this Github repository to explore this process in 1D and 2D using techniques from plasma physics.

2 1D Toy Model

Assumptions

For initial exploration of the plasma, we first formulate a 1D model which will effectively capture the qualitative behavior of plasma wakefield acceleration. We can do this with the following assumptions as they maintain the final derived plasma frequency in our analytic equations:

- The plasma is 1D ($\nabla \Rightarrow \frac{\partial}{\partial x}$)
- The plasma is a cold fluid ($p = 0$)
- The plasma has immobile ions ($n_i = n_0$ constant, $v_i = 0$)
- The plasma is electrostatic ($E = -\frac{d\phi}{dx}$)
- The plasma is affected by no magnetic fields ($B = 0$)
- The plasma is non-relativistic
- The plasma is quasistatic (explained in depth later in the report)
- The plasma has a linear response to perturbations ($n = n_0 + n_1$, $v = v_0 + v_1$, $E = E_0 + E_1$)

Harmonic Oscillation

Using all these assumptions, we create a model for our plasma based on solving the continuity and momentum equations for a plasma and Poisson's equation

$$\text{Continuity:} \quad \frac{\partial n_1}{\partial t} + n_0 \frac{\partial v_1}{\partial z} = 0$$

$$\text{Momentum:} \quad m_e \frac{\partial v_1}{\partial t} = -eE$$

$$\text{Poisson:} \quad \frac{\partial E}{\partial z} = -\frac{e}{\varepsilon_0}(n_1 + n_b)$$

where n_b is the term representing density of the prescrived drive beam. We can proceed to derive an equation showing harmonic oscillation in the linear response of n_1 and E as follows:

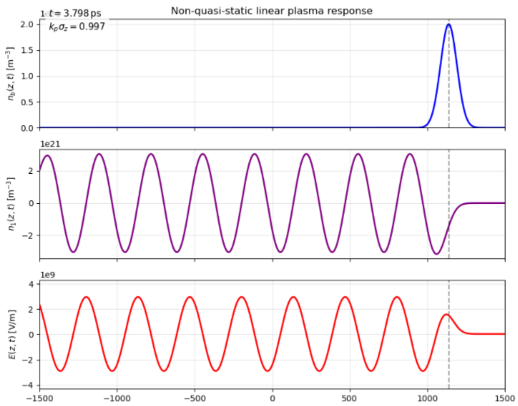

Figure 1: Graphs of $n_b(z,t)$, $n_1(z,t)$, and $E(z,t)$. You can observe oscillatory behavior in n_1 and E once n_b is applied. n_b is a Gaussian.

$$\frac{\partial}{\partial t}\left(\frac{\partial n_1}{\partial t} + n_0 \frac{\partial v_1}{\partial z}\right) = 0$$

$$\frac{\partial^2 n_1}{\partial t^2} - \frac{e n_0}{m_e}\frac{\partial E}{\partial z} = 0$$

$$\frac{\partial^2 n_1}{\partial t^2} + \underbrace{\frac{n_0 e^2}{m_e \varepsilon_0}}_{:=\omega_p^2}(n_1 + n_b) = 0$$

$$\boxed{\frac{\partial^2 n_1}{\partial t^2} + \omega_p^2 n_1 = -\omega_p^2 n_b}$$

Numerical results are shown in Figure 1.

Quasistatic Assumption

The quasistatic assumption encapsulates the idea that the wakefields appear frozen in place relative to the drive beam. This allows us to remove time-dependence from the equation ($n_b(z,t) = n_b(\xi)$). This can be shown analytically with the following coordinate transformation, also yielding oscillatory motion:

$$\boxed{\xi = z - ct, \qquad \frac{\partial}{\partial t} \to -c\frac{d}{d\xi}, \qquad \frac{\partial}{\partial z} \to \frac{d}{d\xi}}$$

$$\frac{d^2 n_1}{d\xi^2} + \underbrace{\frac{\omega_p^2}{c^2}}_{:=k_p^2} n_1 = -k_p^2 n_b$$

$$\frac{dE}{d\xi} = -\frac{e}{\varepsilon_0}(n_1 + n_b)$$

$$n_1 = -\frac{\varepsilon_0}{e}\frac{dE}{d\xi} - n_b$$

$$-\frac{\varepsilon_0}{e}\frac{d^3E}{d\xi^3} - \frac{d^2 n_b}{d\xi^2} - k_p^2\frac{\varepsilon_0}{e}\frac{dE}{d\xi} - k_p^2 n_b = -k_p^2 n_b$$

$$\boxed{\frac{d}{d\xi}\left(\frac{d^2E}{d\xi^2} + k_p^2 E\right) = -\frac{e}{\varepsilon_0}\frac{d^2 n_b}{d\xi^2}}$$

We make two assumptions before simulation. Firstly, the velocity of the beam is equivalent to the speed of light which is valid because the electrons take very little energy to accelerate to this point. Secondly, the beam density does not deform as it travels through the plasma. This means we are not accounting for feedback and beam-loading effects. Our simulations solve for E in the integrated equation

$$\boxed{\frac{d^2E}{d\xi^2} + k_p^2 E = -\frac{e}{\varepsilon_0}\frac{dn_b}{d\xi}}$$

The electric field equation shows that the longitudinal wakefields also act like harmonic oscillators, but its forcing term is the spatial derivative of the drive beam.

To numerically solve this ODE, we used the causal Green's function

$$G(\xi, \xi') = \begin{cases} \frac{1}{k_p}\sin(k_p(\xi - \xi')) & \text{if } \xi' \leq \xi \\ 0 & \text{if } \xi' > \xi \end{cases}$$

and solved the integral

$$E_z(\xi) = \int_{\xi}^{\infty} G(\xi, \xi') f(\xi')\, d\xi'$$

where $G(\xi, \xi')$ is the Green's function at our source and $f(\xi')$ is the source term $-\frac{e}{\varepsilon_0}\frac{dn_b}{d\xi'}$. Since n_b is Gaussian, this integral becomes that of a sine times the derivative of a Gaussian which yields

$$E_z(\xi) = -\frac{en_b}{\varepsilon_0 k_p}\int_{\xi}^{\infty} \sin(k_p(\xi - \xi'))\frac{d}{d\xi'}\left[\exp\left(-\frac{\xi'^2}{2\sigma_z^2}\right)\right] d\xi'$$

Numerical results are shown in Figure 2.

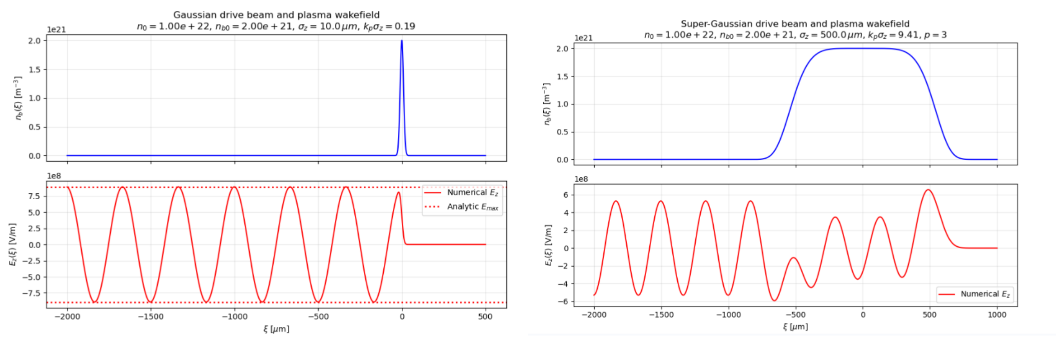

Figure 2: The electric fields $E(\xi)$ with respect to a Gaussian and super-Gaussian drive beam.

Toy Model Results

Using our model, we are able to observe a number of qualitative results from differing drive beams. We first looked at how the maximum amplitude of the wake is related to the width of the drive beam. We scanned through several beam widths and plotted the corresponding maximum wakefield (Figure 3). We found that when total charge is kept constant, wake amplitude increases with decreasing beam width. When the peak density is kept constant, their is a nontrivial optimal beam width. These relations can be derived analytically by using the Greens function solution above and taking $\xi \to \infty$:

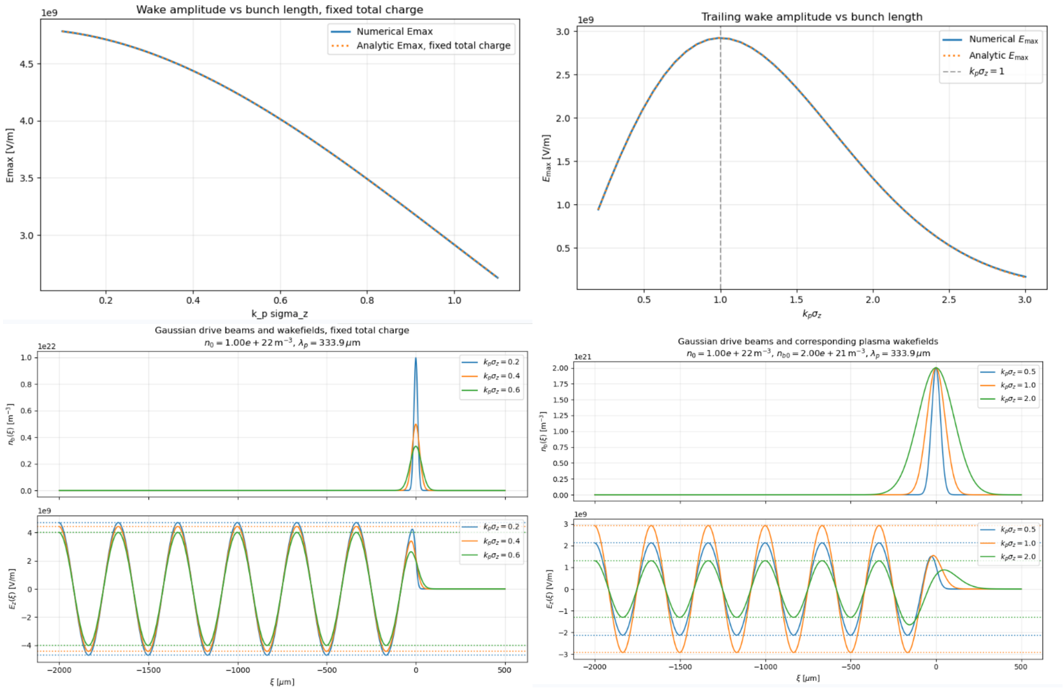

Figure 3: Beam width vs Wake amplitude. Left: constant total charge. Right: Constant peak charge.

$$E_z(\xi) = -\frac{en_b}{\varepsilon_0 k_p}\int_{-\infty}^{\infty}\sin(k_p(\xi-\xi'))\frac{d}{d\xi'}\left[\exp\left(-\frac{\xi'^2}{2\sigma_z^2}\right)\right]d\xi' = \sqrt{2\pi}\frac{en_b}{\varepsilon_0}\sigma_z e^{-k_p^2\sigma_z^2/2}\cos(k_p\xi)$$

There is a total energy gain of

$$\boxed{\frac{\Delta\mathscr{E}}{L} = \frac{q}{L}\int_0^L E_z(z)\,dz = qE_{\text{eff}} \approx 1\text{ GeV/m}}$$

which is 10 times stronger than a traditional RF cavity.

3 2D WarpX Simulations

What is WarpX?

We continue our exploration of PWFA by extending our model by using WarpX. WarpX is a very popular Particle-In-Cell code from Los Alamos National Laboratory used to simulate a variety of phenomena from plasma physics. It includes some PWFA examples which we used to validate the phenomena we found in the 1D toy model. In addition to extending our model to 2D, WarpX updates particals fully consistent with all of Maxwells equations and relativistic effects. This means this simulations account for relativity and beam-loading.

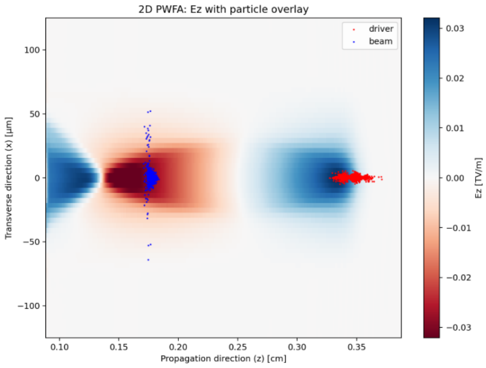

Figure 4: A graph of the longitudinal electric field of a 2D WarpX simulation. Run `plot_pwfa.py` if interested in the results changing beam width.

Results

Using WarpX, we are able to run simulations and plot useful results. By plotting the longitudinal electric field, we observe oscillatory behavior of the electric field, validating the oscillatory behavior of our 1D model. There is also a clear bubble which reflects the earlier description of PWFA. Plotting multiple beam widths using WarpX, we are able to further validate our 1D model by plotting the amplitudes of the beams.

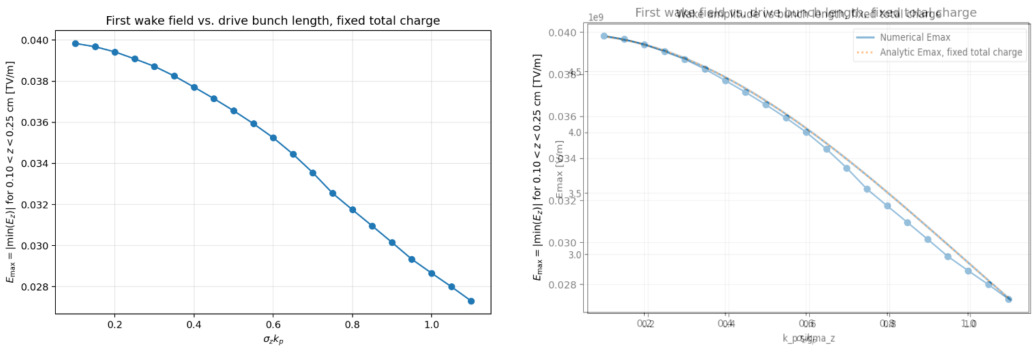

Figure 5: Wakefield vs drive bunch length in 2D and overlayed with 1D.

And as we set out to do, we see not only a decreasing trend, but an extremely good fit to the same graph for our 1D model, validating it even further.